

KAIMING LIU

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RESEARCH INTERESTS

Scientific machine learning; inverse problems; probabilistic inference;
simulation-to-experiment transfer; computational condensed matter physics.

EDUCATION

[University of Illinois Urbana-Champaign](#)

Ph.D. in Physics, in progress

Aug 2023–present

Advisor: Bryan K. Clark

[Xi'an Jiaotong University](#)

B.S. in Physics (Honors)

Sep 2019–Jul 2023

[University of California, Berkeley](#)

Berkeley Physics International Education (BPIE) Program

Aug–Dec 2021

Non-degree visiting program

[Xi'an Jiaotong University](#)

Special Class for the Gifted Young

Sep 2017–Jul 2019

RESEARCH EXPERIENCE

[University of Illinois Urbana-Champaign](#)

Graduate research, Bryan K. Clark group

Nov 2025–present

- Develop probabilistic and multimodal inference for crystal electric field inverse problems, with physics-based forward reconstruction of observables.
- Study robustness to experimental noise and simulation-to-experiment distribution shift in magnetic and scattering inverse problems.
- Implemented neural-network backflow wavefunctions and variational Monte Carlo for correlated fermions, with parallel cluster sampling and reuse of wavefunction evaluations.

[University of Illinois Urbana-Champaign](#)

Research assistant, Vidya Madhavan group

Aug 2023–Oct 2025

- Grew thin-film heterostructures by molecular beam epitaxy and characterized materials with STM/STS in ultrahigh vacuum.
- Automated MBE temperature control with feedback, measurement logging, and alerts for repeatable growth protocols.
- Developed Python and MATLAB STM/STS analysis, including constant-energy contours and joint-density-of-states calculations for quasiparticle interference.

PUBLICATIONS

[P1] Evidence of intertwined pair density and charge density wave orders in UTe₂

Zhen Zhu, Yudi Huang, Julian May-Mann, **Kaiming Liu**, Zheyu Wu, Shanta R. Saha, Johnpierre Paglione, Alexander G. Eaton, Andrej Cabala, Michal Vališka, Eduardo Fradkin, Vidya Madhavan.

PNAS 123 (29), e2602117123 (2026). [arXiv](#).

PREVIOUS RESEARCH	Xi'an Jiaotong University	
	Research assistant, Yongchang Zhang group	Mar 2022–Jul 2023
	- Simulated Rydberg-atom dynamics and photon-mediated interactions with MATLAB and Python to study quantum nonlinear optics. - Built a discrete-time lattice simulation to examine two-qubit phase-flip dynamics and explore many-body regimes.	
	University of California, Berkeley	
	Research assistant, Michael F. Crommie group	Oct–Dec 2021
	Assisted with synthesis and STM characterization of graphene nanoribbons containing five-membered rings, supporting studies of their electronic states.	
TEACHING	University of Illinois Urbana-Champaign	
	Teaching assistant, PHYS 102	Aug–Dec 2023
SELECTED HONORS	China National Scholarship	2021
	Xi'an Jiaotong University	
	Mathematical Contest in Modeling, Meritorious Winner COMAP	2021
TECHNICAL SKILLS	<i>Programming and computing:</i> Python, NumPy, SciPy, JAX, MATLAB; Git; Linux/bash; parallel cluster computation. <i>Scientific methods:</i> Probabilistic inference, Monte Carlo (MCMC/VMC), numerical optimization, forward-model validation, runtime profiling. <i>Experiment:</i> Molecular beam epitaxy, STM/STS, ultrahigh-vacuum systems, feedback control, scientific data analysis.	